

Data Management Plan

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Contributors

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Project

This data management plan describes the data and work in the following project.

SARS-CoV-2 genomic surveillance in Italy

RNA sequencing of nasopharyngeal swab samples taken at a hospital in Italy as part of genomic surveillance of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2).

Start date	2026-11-02
End date	2027-05-03
Funding	: grant number not yet given (applied)

Section A: Description of data – reuse of existing data and/or production of new data

1. How will data be collected, created or reused?

New dataset(s) or AI models

We will create the following new dataset(s) / AI model(s):

- **RNAseq** (Measured data): RNA sequencing of 91 nasopharyngeal swab samples.

This dataset will be produced by a specialized infrastructure: NGL.

For this dataset, the following instrument(s) are used:

- **Illumina NextSeq 500** – The Illumina NextSeq 500 is a benchtop high-throughput sequencing machine developed by Illumina.

Re-used dataset(s)

We will use the following reference dataset(s):

- **genome of SARS-CoV-2 from isolate Wuhan-Hu-1** (Reference data) (https://www.ncbi.nlm.nih.gov/nuccore/NC_045512.2)

2. What types of data will be created and/or collected, in terms of data format and amount/volume of data?

We will be using the following data formats and types:

- **[FASTQ Sequence and Sequence Quality Format](#)**
 - We expect to have between 10 GB to 100GB of data in this format. We use this format because it is a standardised format.
- **[Tab-separated values](#)**
 - We expect to have less than 10 GB of data in this format. We use this format because it is a standardised format.

Section B: Documentation and data quality

3. How will the material be documented and described?

We will use an electronic lab notebook (ELN) to make sure that there is good provenance of the collected data.

In order to keep track of what has been run during analysis, we will:

- Use software tools that are professionally maintained, with version control
- Use standard workflow engines and automatic workflows
- Use interactive notebooks

We will use the following metadata standards to describe our datasets:

- [ENA Sample Checklists](#)

In addition, the following metadata will be provided as documentation: File organisation and naming conventions will be documented in a README text file that will be put in the root folder of the project. Also, a dictionary of the sample metadata will be created.

Data organisation and naming conventions

We have a documented strategy for file and folder organisation.

There will be a README file in the top folder, describing the folder structure and naming conventions.

4. How will data quality be safeguarded?

RNAseq

Regarding the instrument(s) used to generate the data:

- [Illumina NextSeq 500](#) is very well known and described.

Quality processes to be applied include:

- Repeated measurements
- Instrument calibration

Additional quality processes are: A dictionary of the sample metadata will be created.

Section C: Storage and Backup

5. How is storage and backup of data and metadata safeguarded during the research process?

We have consulted institutional functions in order to ensure that we fulfill the policies and expectations of our research institute.

All project data and metadata will be in backed-up storage systems during the project.

All essential data is also stored elsewhere to prevent a total loss of data: The raw reads will be submitted to ENA upon arrival.

The project data will be shared using the following work space service:

- NAISS center

6. How is data security and controlled access to data safeguarded?

In the shared work space(s), only project members have read/write access to the data. Project members will not store data or software on computers in the lab or external hard drives connected to those computers. They can carry data with them on encrypted data carriers and password-protected laptops. Project members have been instructed about both generic and specific risks to the project. There is only a small risk of information loss, and leak. There is an acceptably low risk of information vandalism.

Section D: Legal and ethical aspects

7. How is data handling according to legal requirements safeguarded?

Data we reuse

For the (non-)reference dataset that we reuse, condition is as follows:

- **genome of SARS-CoV-2 from isolate Wuhan-Hu-1** (Reference data) – freely available for any use (public domain or CC0).

Data we collect

None of the data we collect constitutes personal data under GDPR.

Other legal considerations

All data will be owned by the university.

8. How is correct data handling according to ethical aspects safeguarded

Section E: Accessibility and long-term storage

9. How, when and where will research data or information about data (metadata) be made accessible?

We will be working according to the philosophy *as open as possible* for our data. Some of our data cannot be openly shared directly after production:

- Papers needs to be submitted first.

Data that is not legally restrained will be released after a fixed time period: 1 year.

The data / AI models will be accessible as follows:

- **RNAseq** (RNA sequencing of 91 nasopharyngeal swab samples)
 - Published in a domain-specific repository: [European Nucleotide Archive](#).
 - The dataset will be available under the following license: Freely available for any use (public domain or CC0).
 - The data will be published in a repository that is the prime source of data for re-use in the field.

10. In what way is long-term storage safeguarded, and by whom?

The long-term plan for the produced data is as follows:

- **RNAseq** (published in [European Nucleotide Archive](#))
 - The dataset will be stored using storage provided by the institute.
 - This dataset will be kept available as long as technically possible.
 - The metadata will be available even when the data no longer exists.

11. Will specific systems, software, source code or other types of services be necessary in order to understand, partake of or use/analyse data in the long term?

Long-term suitability of the data formats used:

- **[FASTQ Sequence and Sequence Quality Format](#)**
This is a suitable format for long-term storage. No special software or tool is needed to use this material.
- **[Tab-separated values](#)**

This is a suitable format for long-term storage. No special software or tool is needed to use this material.

12. How will the use of unique and persistent identifiers be safeguarded?

- Dataset /AI model **RNAseq** will have a persistent identifier.

Section F: Responsibilities and Resources

13. Who will be responsible for data management?

Yvonne Kallberg is responsible for overall data management including implementing the DMP, and ensuring it is reviewed and revised.

14. What resources will be required for data management?

To execute the DMP, additional specialist expertise is required which will be accomplished by:

- Training existing staff: Yvonne Kallberg will attend the NBIS course 'Introduction to data management practices'.

None of the repositories or long-term storage providers used charge for their services.